

Adult Income Prediction - ML Foundations

Problem

Predict whether an individual earns more than \ per year (1 = >50K, 0 = <=50K).

Business Objective

Identify people likely to earn above \ so outreach can focus on promising customers instead of spending effort on lower-income individuals.

Primary Metric

F1 score was selected as the main evaluation metric. Precision alone would reduce wasted outreach but miss many actual high-income people. Recall alone would find more of them but increase false targets. F1 balances both types of error.

Dataset

Rows: 48,842 | Features: 14 | Positive class: 23.93% | Negative class: 76.07%

Age centers around the late 30s, hours-per-week clusters near 40, capital-gain is heavily skewed toward zero, and HS-grad / Bachelors are the most common education levels.

Baseline Results

Majority Baseline - Accuracy: 0.7607, Precision: 0.0000, Recall: 0.0000, F1: 0.0000, ROC AUC: 0.5000, PR AUC: 0.2393

Education Rule (edu-num >= 13) - Accuracy: 0.7530, Precision: 0.4844, Recall: 0.4970, F1: 0.4906, ROC AUC: 0.6653, PR AUC: 0.3611

The education rule performs better on F1 (0.49 vs 0.00). A stronger model would need to improve clearly on that score.

Error Analysis

False Positives (1,237): Many have Bachelors-level education but still earn <=50K, often in lower-paying roles such as Adm-clerical or Other-service.

False Negatives (1,176): Several high-income individuals have education-num below 13 but show other signals such as higher capital-gain, longer hours, or professional occupations.

Data Issues Identified

1. Missing values in workclass, occupation, and native-country
2. Categorical columns need encoding before modeling
3. Skewed numeric features (capital-gain, capital-loss)
4. Class imbalance (~24% positive)
5. Single-feature rule is too limited
6. Feature interactions (education + occupation) are not captured

Conclusion

F1 will remain the primary metric. The majority baseline reached F1 = 0.00, while the education rule reached F1 = 0.4906. Further modeling work will aim to improve on that baseline while keeping precision and recall at reasonable levels.